Mehdi Eskandari-Ghadi

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EDUCATION

Civil Engineering Ph.D. (GPA 4.0) **Boulder, CO**
Engineering Science Focus 2016 – 2021
University of Colorado Boulder

TECHNICAL SKILLS

Programming	Python, TypeScript/JavaScript, Git, VBA, MATLAB, AutoLISP
Platforms & Data	Palantir Foundry (Ontology, Pipeline Builder, Contour, Workshop, OSDK, AIP), Spark, Polars, Pandas
Engineering	AutoCAD, PLSCADD, L-PILE, ABAQUS/LS-DYNA
Standards & Tools	NESC, GO-95, FERC-881, Microsoft Office Suite, LaTeX
Certifications	Palantir Foundry & AIP Builder Foundations; Data Analysis in Contour (Deep Dive); Your First AIP Workflow, Your First End-to-End Workflow, and Data Science Fundamentals (Speedruns)

ENGINEERING EXPERIENCE

System Modeling Principal Engineer (Palantir Foundry) — PacifiCorp **Jan 2025 – Present**

- Designed, built, and deployed the company's Foundry-based facility rating data management system end to end with full-stack TypeScript and Python, migrating prototype FERC-881 line-rating and FAC-008 equipment-rating business logic into production. The system spans backend and queue-based processing pipelines, user-facing applications, audit preparation, and interdepartmental data collection, delivered concurrently with testing and production adoption.
- Built an enterprise, operations-scale Foundry ecosystem on an Ontology redesigned for semantic clarity and workflow object types, with 100+ Actions and Functions and 30+ Pipeline Builder pipelines. It manages transmission facility and equipment records and their modifications, computes granular rating calculations, and interfaces with external systems through automations, map interfaces, graphs, and other UI elements. Owned end-to-end UX design in Foundry Workshop, continuously identifying friction points and iterating on layout and information architecture to keep complex, multi-source engineering workflows simple and intuitive for end users, with shared layouts, color schemes, and reusable components (map, facility table, metric cards) across all modules. Drove adoption by training PacifiCorp engineers on the system, writing user guides and code documentation, and holding weekly feature-update meetings.
- Directed AI coding agents with repository and Foundry Ontology access by sequencing incremental implementations and large migrations to write and edit functions, entire features, and React-based custom Workshop widgets through a full commit, tag, and pull-request workflow, coordinating agent-driven changes across multiple interdependent repositories while maintaining version and compatibility consistency.
- Built a reusable Python report-processing library and used it to power LLM-based reporting via Palantir AIP, generating narrative reports on system modification changes and their system implications, in active use to focus administrative review across data spread over legacy tables, files, engineering drawings, and internal apps.
- Architected a reusable, git-inspired version-control framework for system analysis lifecycle management in Foundry: facility analyses are cloned, modified, and published in stages through department-specific review checkpoints, with automated resource tracking and validation.
- Built reliability safeguards: unit and end-to-end tests, dataset primary key checks, validation rules, monitoring panels flagging missing object links, and pre-run validation previews for resource-consuming tasks.
- Designed backend Python data pipelines in Spark and Polars to ingest, parse, and remediate enterprise-scale, imperfect legacy data from multiple database and spreadsheet sources, automating cleanup and multi-source merging to reconcile records to production quality standards.
- Developed a reusable Python library implementing physics-based, IEEE-compliant thermal rating models, numerically solving the nonlinear ordinary-differential heat balance (solar heat gain, convective cooling, radiative heat loss, conductor resistance) through Runge-Kutta forward integration and Newton-Raphson inverse root-finding, vectorized at scale for static, dynamic, and transient line and equipment ratings within the platform. Implemented seasonal ambient temperature analysis over 30 years of daily weather data across thousands of miles of power lines.
- Designed and built a streamlined, user-friendly FAC-008 substation equipment rating tool from scratch in VBA/Excel, applying IEEE equipment-rating knowledge to compute normal and emergency ratings and rank limiting elements for breakers, switches, capacitors, reactors, buses, jumpers, and transformers from company standards and nameplate specifications.
- Applied geometric and spatial analysis methods to compute mutual-impedance candidate pairs, line-distance statistics, and line-to-line distance visualizations for parallel transmission corridors, integrated into platform UI graphs.

- Deployed AutoCAD experience and problem breakdown skills to implement AutoLISP one-line-drawing parsers. The parsers identify transmission facilities and equipment, deduce line segment and equipment configurations, and export structured data to reconcile database records with engineering drawings.

Transmission Engineer I/II — PacifiCorp **Jan 2023 – Jan 2025**

- Designed new transmission lines and rebuild and maintenance projects for existing lines in six states with PLSCADD, L-PILE, NESC, and GO-95, collaborating with teams of engineers, managers, field crews, and legal, and meeting regularly with external customers to coordinate reroutes for their projects and line changes encroaching on their properties.
- Scoped new transmission lines with other engineering disciplines, and applied structural and geotechnical engineering to custom-engineer transmission structures and drilled pier foundations for special needs, with reference to ASCE, ACI, and FHWA.
- Overhauled the company's transmission structure library from legacy templates into modular, component-based PLS-POLE libraries, and designed a new class of circuit-swap structures for repositioning circuits on a double-circuit extra-high-voltage line with minimal increase to structure size, vetted across engineering and field operations to meet electrical and structural requirements.
- Utilized VBA and Excel programming alongside transmission line rating engineering knowledge to design and build an auditable, automated FERC-881 steady-state line-rating analysis tool from scratch. This tool integrates PLS-CADD structure data, conductor properties, and historical weather data with CIGRE-referenced seasonal rating methodology to compute normal and emergency ratings and critical-span summaries. It streamlined grid line data extraction, reducing engineering hours threefold, and initialized the Foundry datasets for future system migration, validation, and automation.
- Developed automated Excel tools for deep foundation design, construction package generation, transmission pole design aids, and company sag-tension standards for tree-wire.
- Designed and reviewed drawings for tubular steel poles and drilled piers to withstand vertical and lateral loads, and improved company standards and procedures through detail-oriented QAQC reviews.
- Performed detail-oriented technical review of transmission standards, contributing to their development, and initiated process improvement pipelines for transmission design processes and documentation.

RESEARCH EXPERIENCE

Graduate Research Assistant / Ph.D. Candidate / Postdoctoral Researcher **Jun 2018 – Jan 2023**

University of Colorado Boulder, Boulder, CO

- Collaborated with a team of senior researchers at the Lawrence Berkeley National Laboratory to investigate and develop mechanistic models for subcritical crack growth healing.
- Utilized knowledge of fundamentals of fracture mechanics, fluid transport, and boundary element methods to develop software that couples fields of mechanics and physics.
- Merged expertise in research, programming, and engineering to investigate multiphysical and multiscale solid-fluid interactions (e.g., adsorption, surface tension, solvation pressure) in porous solids and develop a thermodynamics-poromechanics-based MATLAB software for "adsorption-induced deformation of microporous material", running parametric studies to test model sensitivity.
- Employed technical writing and illustrative skills to author 4 peer-reviewed publications in respected journals of JMPS, IJSS, and EFM.

EMI Virtual Conference 2021 and EMI Conference 2022 **May 2021 and May 2022**

- Applied illustrative and verbal communication skills to deliver presentations titled "A multi-scale theory explaining the initial shrinkage of micro-porous solid upon gas adsorption" and "A Multiphysical Surface-Force Based Fracture Theory for Subcritical Crack Growth in Surface-Reactive Environments", followed by Q&A from engineering and academic audiences.
- Received runner up award in the 2021 Student Poster Competition for poster presentation on sorption-induced deformation of porous material.

Lawrence Berkeley National Laboratory BES **Sep 2020 – Jan 2023**

- Presented regular scientific presentations on topic of crack propagation to teams of research professionals and senior scientists in the LBNL BES with diverse backgrounds in mechanics, physics, and chemistry.

TEACHING EXPERIENCE

Teaching Assistant / Tutor **2014 – 2019**

University of Colorado Boulder, Boulder, CO & University of Tehran, Tehran, Iran

- Presented class lectures and laboratory teaching sessions, held office hours and exam preparation sessions, prepared and graded exams and homework question sets, for courses of Advanced mechanics of materials and Dynamics.
- Tutored students for FE exam with 100% success rate.

- Tailored teaching approach specific to different student learning styles and utilized CAD skills to enhance visualization and understanding.
- Held office hours and graded homework, quizzes, and class projects for Structural Analysis II and Hydraulics of open channels as a Teaching Assistant at the University of Tehran, Fall 2014 and Fall 2015.

LEADERSHIP ROLES

Research Mentor **Jan 2022 – Jan 2023**

University of Colorado Boulder, Boulder, CO

- Trained new Ph.D. students on fundamentals of mechanics, written and oral skills, and technical research.
- Guided a research trainee through the modeling and simulation of beam bending in humidity actuators.

President/Co-president — Persian Student Organization, Boulder, CO **Aug 2018 – May 2019**

University of Colorado Boulder, Boulder, CO

- Proposed, secured funding, planned, and oversaw student events in collaboration with professionals from industry and academia, local businesses, and non-profit organizations.